Development project in Machine Learning

TAF MCE

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Objectives

Versioning with GIT

Project description

Deliverable

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- Develop good programming practices
- Use standard development tools
- Get used to collaborative work

- Work on Machine-Leaning datasets

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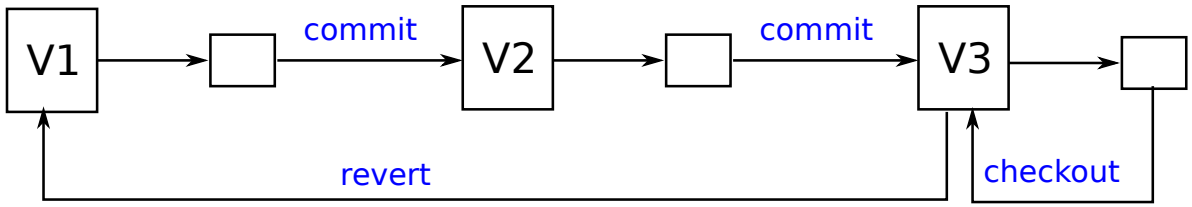
- GIT is a versioning system
- It can keep track of the successive code versions
- It allows several persons to work on the same files, and can merge the various contributions

Tutorials:

- `https://openclassrooms.com/fr/courses/1233741-gerez-vos-codes-source-avec-git` **(Course)**
- `https://github.com/girliemac/a-picture-is-worth-a-1000-words/tree/main/git-purr` **(Pictures)**
- `https://ohmygit.org/` **(Game!)**

How GIT works

- Versioning:

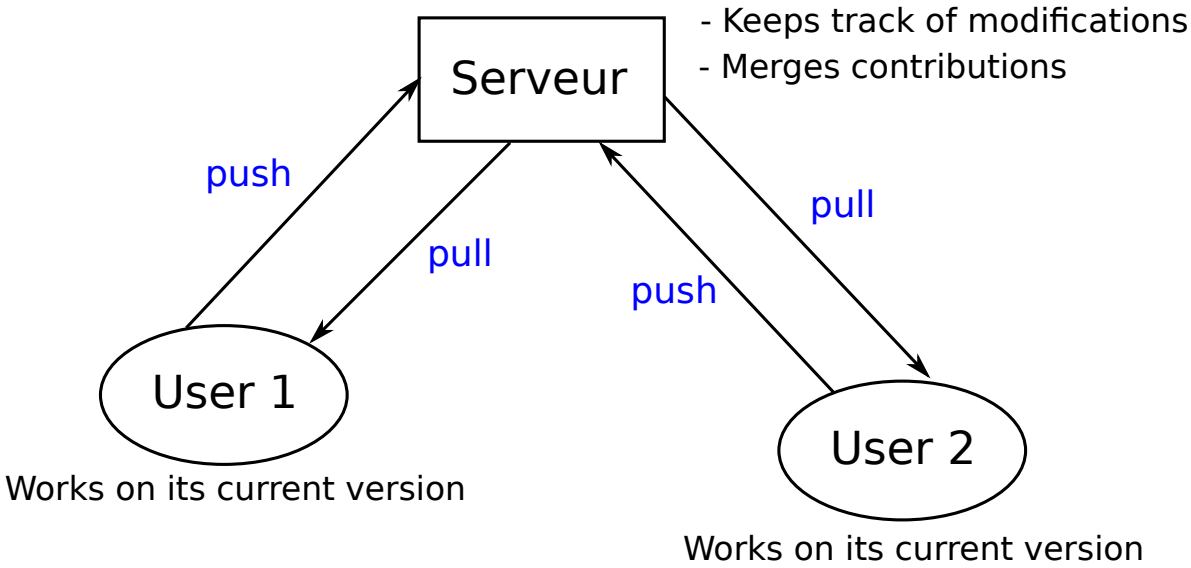


Example: coffee machine

- V1: template (repository organization, readme, etc.)
- V2: Added function to boil down water
- V3: Added function to pour water
- Then: Modified function to boil down water, but broke everything!!

How GIT works

- Architecture:

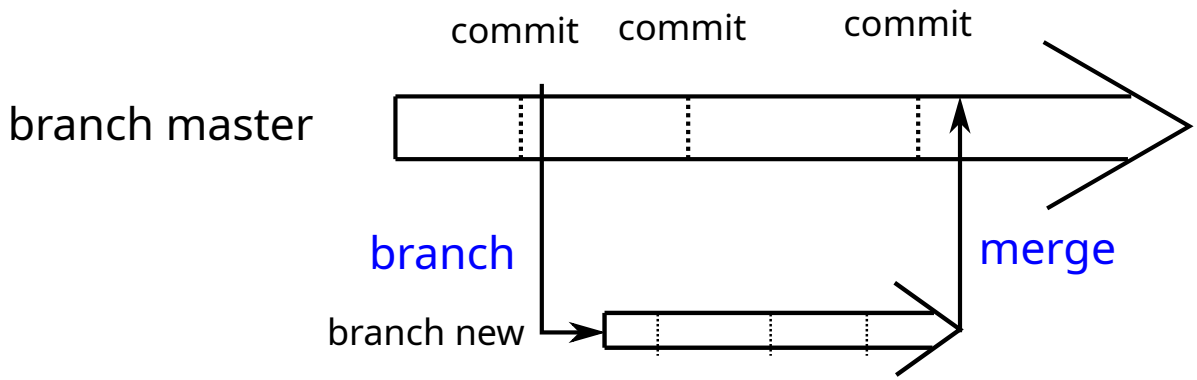


Example: coffee machine

- User1: works on function to boil down water
- User2: works on function to pour water

How GIT works

- Branches:



Example: coffee machine

- branch master: the machine can only makes coffee
- branch new: we want to add modes to choose between coffee or tea

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Machine Learning Workflow

1. Import the dataset
2. Clean the data, perform pre-processing
   - Replace missing values by average or median values
   - Center and normalize the data
   - Apply an PCA
3. Split the dataset into training and testing
4. Train the model, including feature selection
5. Validate the model

Objective: collaboratively implement this workflow and apply it to different ML datasets

Datasets

The objective of the project is to apply Machine Learning methods for Binary classification onto two different datasets (both available on Moodle):

- SPAM Dataset:
  `https://archive.ics.uci.edu/dataset/94/spambase`
  Characteristics: 5000 entries, 50 features, balanced dataset

- Diabetes Health Indicators Dataset:
  `https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset`
  Characteristics: 250k entries, 21 features, unbalanced dataset

Constitute groups of 3 to 4 students.

What I expect (programming)

Main objectives:

1. Work with a git repository for your group: `https://gitlab.imt-atlantique.fr/`
2. Apply the same ML workflow onto the two datasets

In terms of code:

- Write the Python functions implementing the workflow in one single .py file.
- Call the workflow and present the results in two .ipynb files
- Write at least one unit test for one of the functions

What I expect (ML)

- Evaluate and compare a few different methods for classification
- Implement **feature selection**
- Handle specific difficulties of each dataset (missing values, unbalanced dataset, etc.)
- Find relevant performance metrics for the considered problems

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Code: **one .py** file for the functions and **two .ipynb** files to run the code included in the Git repository + all what is needed to run the code (datasets, another .py file for unit tests if needed, etc.)

One Git repository per group should be "sent" by e-mail to `elsa.dupraz@imt-atlantique.fr` before the 21th of November, 11PM.

$\rightarrow$ please give me the sufficient level of rights to see the repository (e.g. developer)

The .ipynb file must...

- Apply the workflow from the functions **of the .py file** onto the two datasets
- Show and compare the results for each dataset (e.g., numbers, curves, etc.)
- Comment on the results in both cases (which method would you choose at the end?)

If you used AI, explain where, how, and why (can be included in the ipynb file, or in the git repository).

Final comments

- Advice for good programming practices:
  `https://mikecroucher.github.io/reproducible_ML/`
- Register your groups before the 10th of October, at
  `https://semestriel.framapad.org/p/s4kbcomn5l-anxw`